The Red Queen 120-cell

Plan: the polar exchange of the White Queen 600-cell (placeholder name)



DICTO · idea 27–28 October 2026: "a polar swap of the White Queen", "can we build the polar exchange 120 cell" · study 2026-10-27-white-queen-polar

> **Goal.** Build the 120-cell, the polar of the 600-cell, exactly, and show the exchange between the two: every vertex of one becomes a cell of the other. The White Queen has 600 tetrahedra round 120 corners; her polar, named here the Red Queen after the other queen of *Through the Looking-Glass*, has 120 dodecahedra round 600 corners. Result wanted: the same proved atlas as #63 for the Red Queen, the exact dictionary between the two, and a Polar button in the White Queen world.

Already proved by our own data

The corners of the Red Queen are exactly the centres of the White Queen's 600 cells. So the Cleo shells of #63 are already a result about the Red Queen:

White Queen 600-cell (#63)	Red Queen 120-cell
600 cells in 15 shells round one vertex: 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20, 20	600 corners in the same 15 layers round one dodecahedral cell
Core: 20 Cleo-Icos filling a regular icosahedron	Core: the 20 corners of the central dodecahedron
120 vertices in 9 layers round a vertex: 1, 12, 20, 12, 30, 12, 20, 12, 1	120 dodecahedra in the same 9 layers round one dodecahedron
The list reads the same backwards (central symmetry)	The same, for the same reason

So the polar exchange of the Cleo core is the dodecahedron itself: twenty Cleo-Icos round a point on one side, twenty corners round a cell on the other. This is the icosahedron–dodecahedron duality of the Platonic solids, one dimension up.

Stages

Stage	Work	Where	Cost
0	Prior art: Coxeter, *Regular Polytopes*; Alicia Boole Stott's 120-cell sections (about 1900); Schlegel diagrams and physical 120-cell models; every source checked listed.	library + web	small
1	Exact source model of the 120-cell from the #63 source model: 600 corners, 1200 edges, 720 pentagons, 120 cells, every cell a regular dodecahedron; checks as for #63.	dicto-big-brother	small, no tokens while it runs
2	Red Queen shells: what each of the 9 dodecahedron layers looks like in perspective, and whether each layer is one kind of piece, as the Cleo shells are.	dicto-big-brother	small
3	Red Queen atlas: slices and four shadows in the four main directions, exact, as #63; the halves of each shadow.	dicto-big-brother	medium
4	The dictionary: each White Queen slice, shadow and shell beside its polar, in one table and one figure.	study	small
5	App: a Polar button in the White Queen world swaps 600-cell and 120-cell in the Shadow, Slice and Shells views; the gold Cleo core turns into the gold central dodecahedron.	Kaleidohedra	medium
6	Paper #64 (placeholder): The Red Queen 120-cell, led by its main theorem, the exact polar dictionary.	paper	medium

Order

Stages 0–2 now: they run on dicto-big-brother and cost few tokens, so they fit the frugal week. Stages 3–6 after the frugal week, after the build pieces (Icosaxes, Lotus Jewel Axes, Unborn Axes), which stay first.

Name: "The Red Queen 120-cell" is a placeholder until DICTO names it. Carroll references stay in the White Queen and Red Queen papers only.